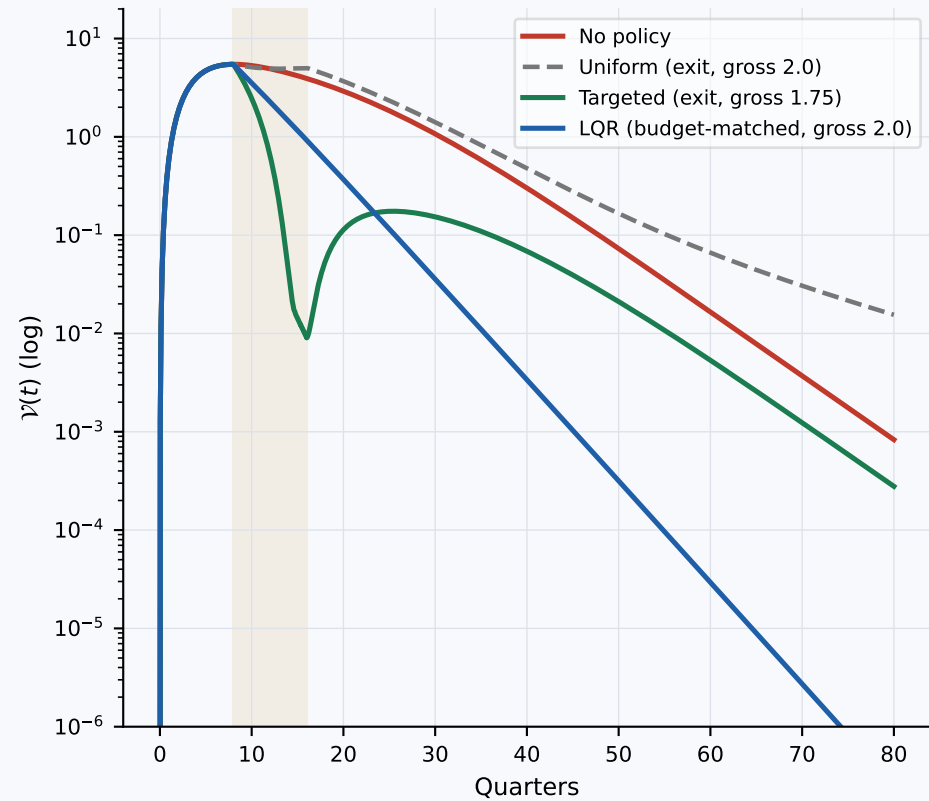
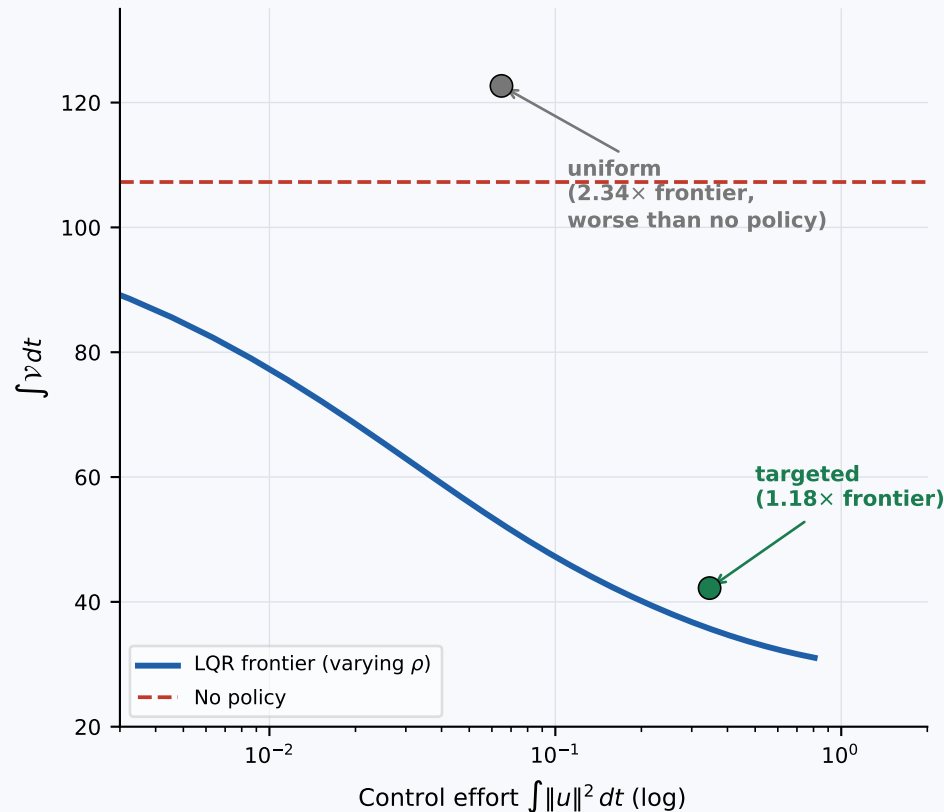


Optimal-Control (LQR) Benchmark: heuristics vs the Riccati feedback $u = -\rho^{-1}Sx$, deployed at $t_0 = 8q$

(A) Lyapunov convergence $\mathcal{V}(t)$



(B) Cost plane: integrated disequilibrium vs control effort



(C) Operational metric: recovery time vs gross expenditure

